Visualization of networks

Pol Monné Parera, Shaubhagya Mahat, Pau Casé Barrera
polmonneparera@lewisu.edu, shaubhagyamahat@lewisu.edu, paucasebarrera@lewisu.edu
DATA-53000, Fall 2025
Data Visualization
Lewis University

I. Introduction

The aim of this project is to investigate different datasets on various scales of mobility networks and how can graph visualizations be used for the study and comparison of these datasets. The transport systems can be modelled as graphs, in which nodes represent physical locations and are interconnected by edges representing the available transportations in the network. The datasets studied focus on three different scales: town level ground transportation network of Kuopio, Finland [1]; global airport traffic [2] as an example of worldwide mobility; and, finally, a country level study of the US domestic airline network [3].

The final objective of the project is to show how graph-based visualization techniques reveal intuitive as well as less obvious characteristics of complex transportation networks on different levels.

II. Dataset Descriptions

. Dataset 1: Kuopio Public Transportation Network [1]

TABLE I. Node Dataset Attributes

Attribute	Type	Example Value	Description
Node ID	Integer	3	Identifier of the node
Latitude	Float	2.340529	Latitude
Longitude	Float	38.23371	Longitude

TABLE II. Edge Dataset Attributes

Attribute	Type	Example Value	Description
Source ID	Integer	3	Identifier of the source node
Destination ID	Integer	5	Identifier of the destination node
Average time to cover edge	Float	3.2	Average time to cover the path from source to destination with public transport

The first network evaluated is the public transportation stops and lines in the city of Kuopio, Finland. This small city allows for easy calculations because of its small size. However, it is situated in a part of the world where lakes are very abundant, and the city is surrounded by water. The thumbnail of the dataset, Figure 1, allows the viewer to see an overall view of the city's connections. Some of the notable takeaways from Figure 1 is that the center of the city seems to be where all the lines meet. There seem to be big neighborhoods in the South and a slightly smaller one in the North. These observations are hypotheses that one can extract from the public transportation lines in the city. Last, there seems to be a separate town that has a public transport line to the East of Kuopio and it is also in the dataset. This image is provided by the dataset and gives context as to what the exploration may result in.

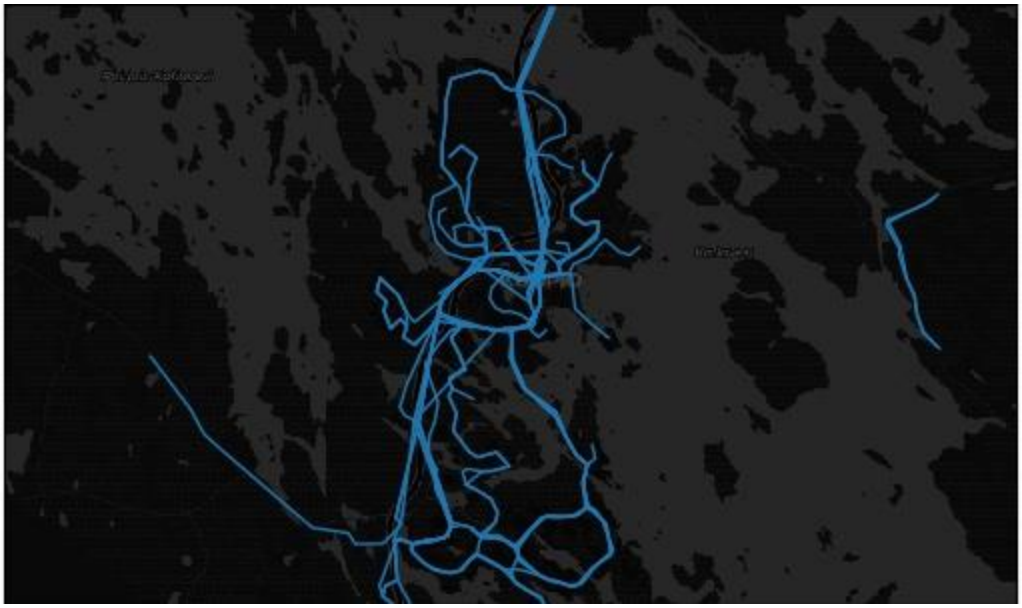

Figure 1: Public transportation lines in Kuopio on top of a map of its geography. Thumbnail of the dataset

The nodes in the graph are the stops, and the edges represent any line that connects two stops. They are weighted depending on the average time to go from one stop to the other.

DATASET 2: WORLDWIDE AIRPORT NETWORK

TABLE I. NODE DATASET ATTRIBUTES

Attribute	Type	Example Value	Description
Node ID	String	MIA	Identifier of the node. Corresponds to the IATA airport code.
Latitude	Float	25.793199539184	Latitude of the airport.
Longitude	Float	-80.29060363769	Longitude of the airport.

TABLE II. EDGE DATASET ATTRIBUTES

Attribute	Type	Example Value	Description
Source	String	MIA	Identifier of the source node. Corresponds to the IATA airport code.
Target	String	BCN	Identifier of the destination node. Corresponds to the IATA airport code.

The second network studied corresponds to the airport traffic network worldwide. Nodes represent specific airports while edges represent directed connections among them. Due to the global scale of the dataset, the network is significantly larger and more complex than the city-level transportation network previously described.

Data for this dataset was extracted from the Open Flights website [2]. The data was preprocessed with Python to prepare it in a format that Gephi software would accept and would allow for work to be done on it, main steps consisted of deleting incomplete nodes and creating a nodes and an edges csv file.

DATASET 3: USA AIRPORT NETWORK

TABLE I. NODE DATASET ATTRIBUTES

Attribute	Type	Example Value	Description
Airport ID	Integer	293	Unique identifier for each airport
Airport Name	String	*Chicago O'Hare International*	Name of the airport
Latitude	Float	41.9786	Geographic latitude of the airport
Longitude	Float	-87.9048	Geographic longitude of the airport

TABLE II. EDGE DATASET ATTRIBUTES

Attribute	Type	Example Value	Description
Source	Integer	11292	Identifier of the departure airport (origin node)
Target	Integer	10397	Identifier of the arrival airport (destination node)

The USA Network of Airline Routes has 235 nodes that represent US major and regional airports, and the 1297 edges indicate the direct domestic flight connections that link these airports. Geographic coordinates (latitude and longitude) are available for all the nodes, which will allow accurate creation of Geographic Layouts in Gephi using the Geo Layout Algorithm. The edges were not weighted to indicate their relative importance; however, they did reflect the actual air transport routes in the United States. The USA Network of Airline Routes has the potential for network analysis using the PageRank Algorithm, Modularity, Degree Centrality, and Betweenness Centrality to identify critical hubs, regional groups and connectivity among airports in the U.S. Air Travel System.

III. METHODOLOGY AND RESULTS. DATASET 1

The density of the graph calculated by Gephi is 0.002, which is very logical considering that most stops have only one public transportation line going through them. The network has a diameter of 62, but this is not very informative. If we remove all stops that are only connected to one line, the diameter is reduced to 12, and for stops with 3 lines, it reduces further to 5. From this information, it can be speculated that some lines connect outer suburbs to the inside of the city, with a lot of stops on the way. As we reduce the number of stops in the suburbs, we reduce the diameter of the network, displaying how in the center, everything is connected through less stops. This may be true for many cities in the world, but it is still good to use this data to understand our transportation network.

Figure 2 displays the graph with the nodes laid out geographically. The red nodes are part of only one bus line (have an out and in degree of 1). The figure uses a divergent color map so that red shows stops with only one transportation line, while darker green shows the stops with the most outgoing edges. In the middle, the gray color displays nodes with medium values for outgoing edges. The minimum value is 1 (red) for outgoing edges and 8 (darkest green) is the maximum.

From this figure, we can also see that the Eastern part of the map has some connections that are not connected to the center of the city, perhaps belonging to a town nearby under the same public transport administration, which can be confirmed by seeing the map in Figure 1, in the description of the dataset.

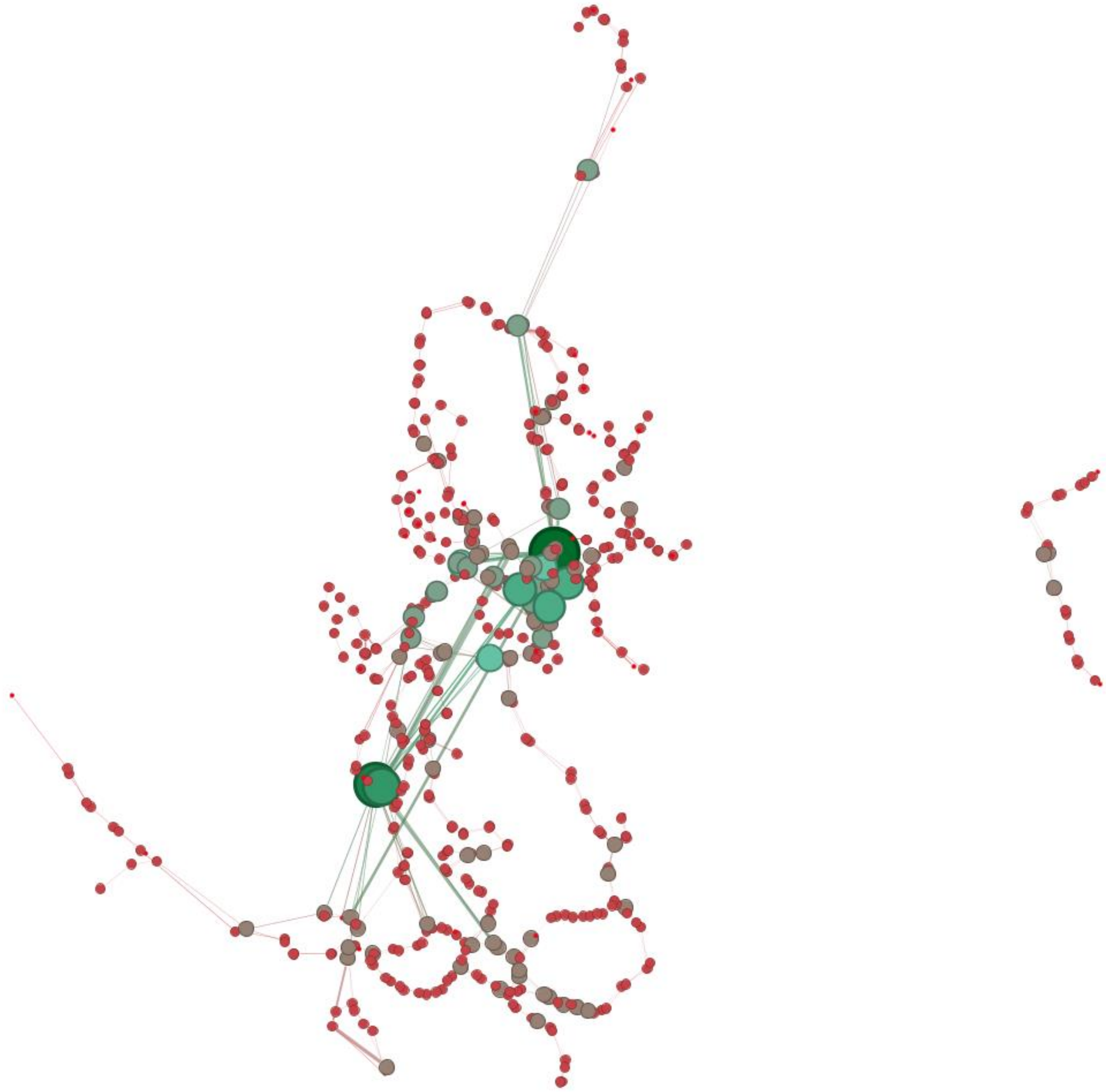

Figure 2: Public transport stops in Kuopio and their connections. Colors indicate number of outgoing connections

Figure 2 also displays the size of nodes depending on their hubness score of the HITS algorithm. This comparison is done despite HITS being built specifically for website ranking, so that we can experiment with the algorithm in this transportation network. We can see the larger nodes usually have darker green tones which reflect the most outgoing edges, denoting a correlation between hubness score and outgoing nodes.

This experiment is used in the report because we can see that the larger nodes tend to also be the ones with the most outgoing edges, and the smaller nodes tend to be the ones with the least edges going out. This is logical considering that hubs connect with other important nodes (authorities). In this graph this is true for the most important nodes in the city center.

The edges are kept on Figure 2 because they allow to see the connections between the stops, even if they do not give specific meaning, they allow to see, for example, that the Eastern town is not connected to the Kuopio city center.

Figure 3 shows the same graph. Now, the nodes use the Modularity feature of Gephi that separates the nodes into different clusters. Only the top 7 clusters are colored, the rest of nodes are in grey, to minimize the confusion that similar colors could give. For the Modularity calculation, a resolution of 6 was used to form larger, more interpretable clusters.

The coloring of only the larger clusters was done considering the number of nodes per cluster. By looking at Figure 4, it is clear that coloring all of the communities will create a distraction with many colors and very small clusters. The figure shows the number of nodes on the Y axis for each of the communities.

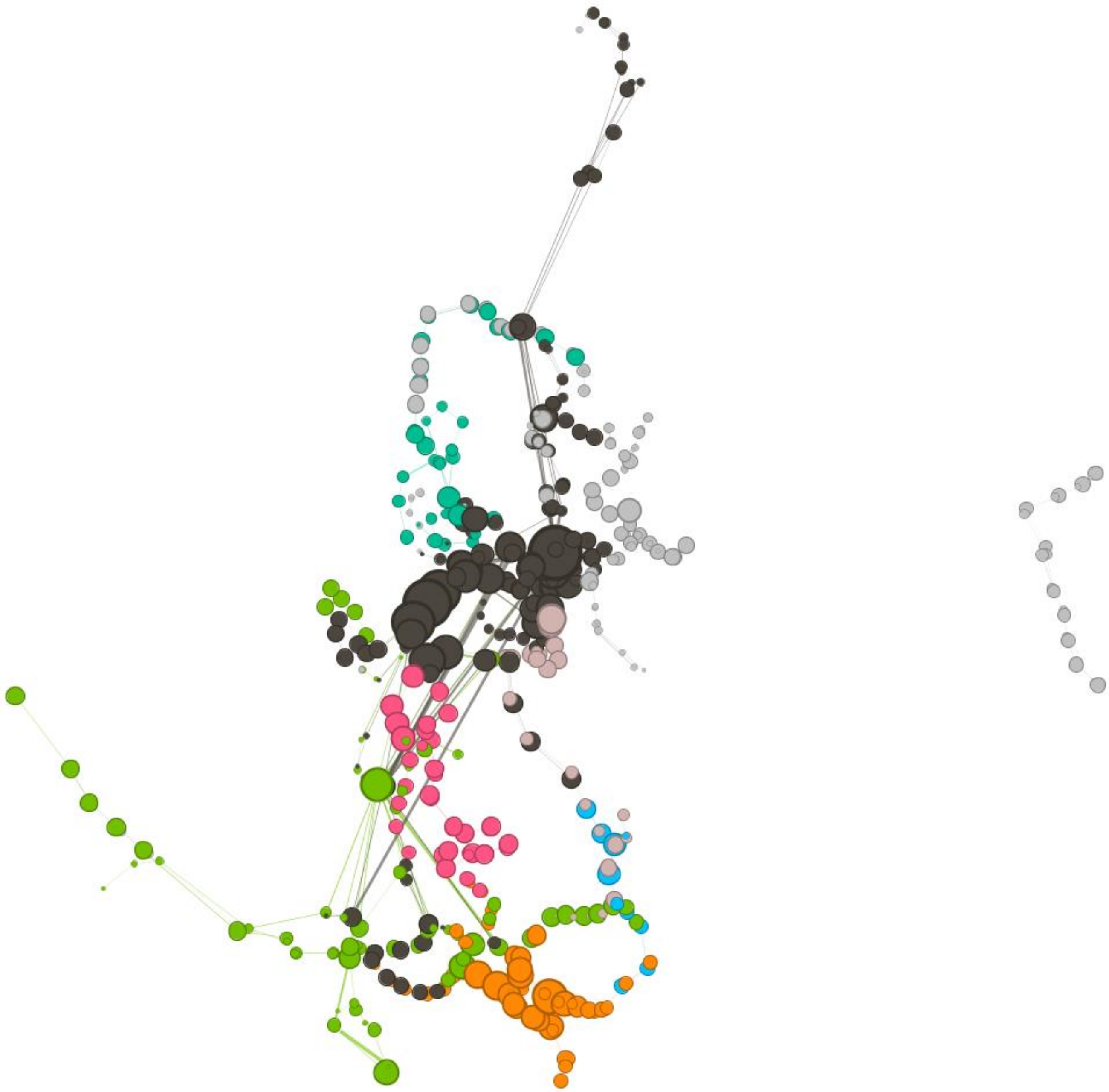

Figure 3: Kuopio Public Transport Network Communities using Modularity and size indicates PageRank

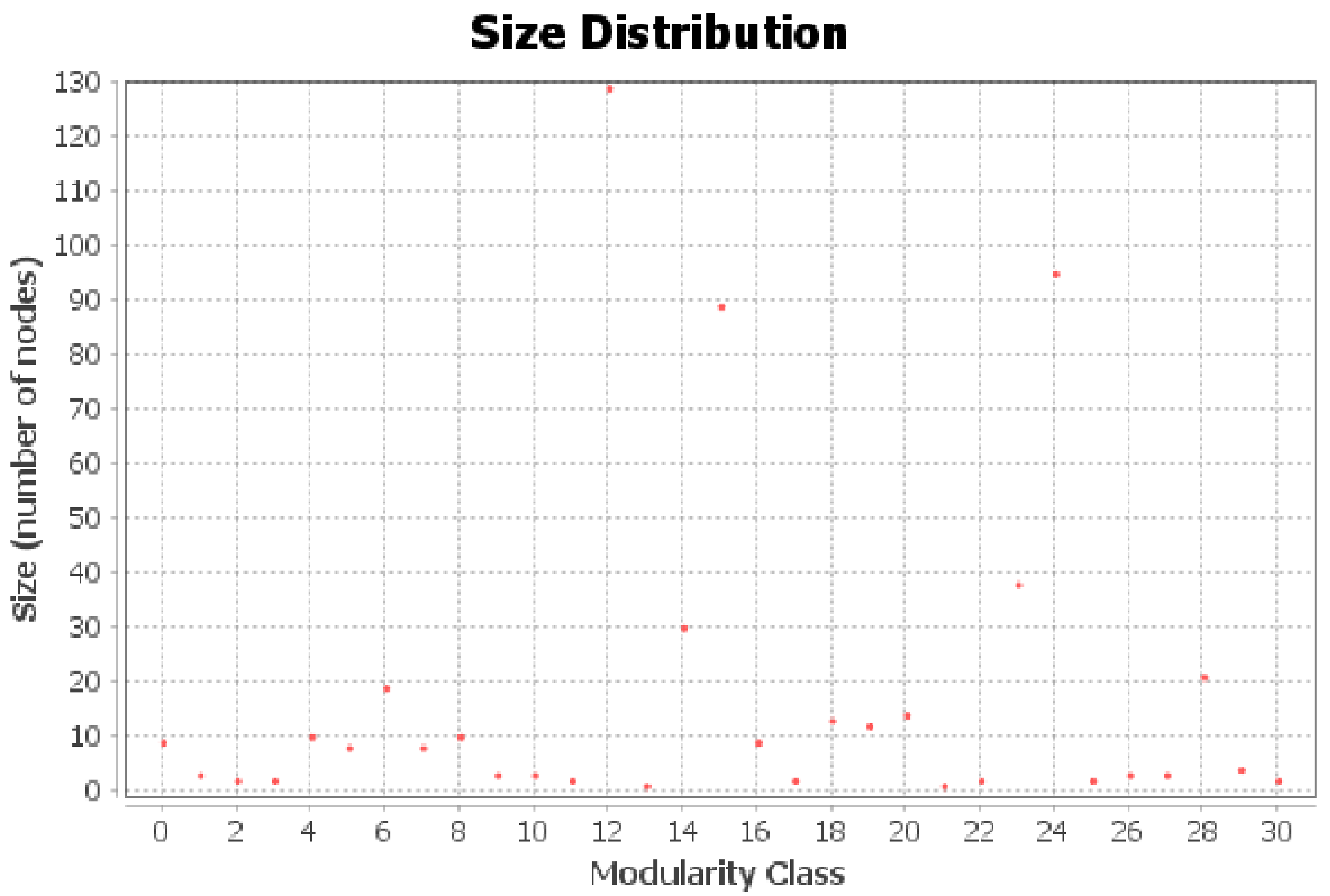

Figure 4: Number of nodes in each community of the Kuopio Public Transport Network

From Figure 3, we can see the connections within the graph form communities around nearby zones, which is logical considering bus stops of a single line are commonly in one part of the city only. However, we can see the black cluster extends into different areas, since the center cluster is connected to most areas of the city. In Figure 3, the size of the circles also denotes the PageRank, which we can see gives a higher score for the centered nodes of each community compared to the ones on the outside of the city center. Additionally, there is a higher PageRank in the middle community thanks to it connecting to many important nodes.

Figure 5 closes the main geographical analysis on the first dataset by showing the betweenness centrality of the nodes in the network in the size of the nodes. The color indicates the degree of the outgoing edges of the node, the darker being the highest.

We can see two very large nodes. The largest one is in the city center, where there are many other transportation stops. The second larger point seems like a transportation hub that connects the city center with the Southern suburbs, which is seen in the edges that connect to the node. This is also seen in Figure 2, but it is not as clear as in this betweenness centrality visualization.

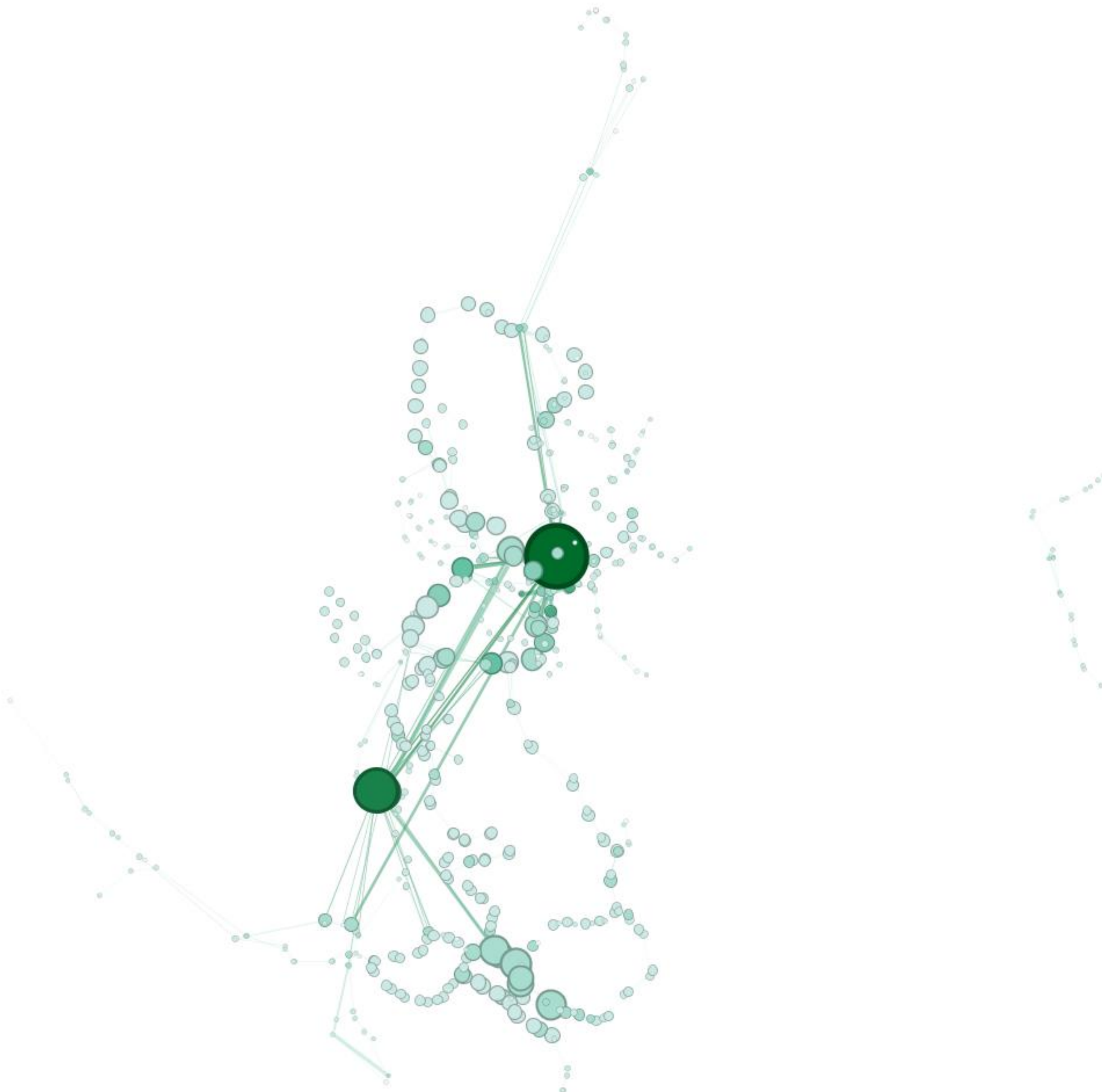

Figure 5: Betweenness Centrality of the nodes in the network in the size of the nodes

Because of the geographic nature of the map, the Geographical Layout plugin [4] was installed into Gephi to be able to show the real distance between nodes, since the analysis done relies on these nodes being public transport nodes and therefore the conclusions that are extracted from the images can be more precise and wider if the nodes are placed in their correct using latitude and longitude. However, since the assignment encouraged us to use other layout algorithms, the following analysis was made.

A complementary analysis has been done to test several layouts that Gephi provides. Despite this being a geographical dataset, we can try to study it from a geographically agnostic network point of view. Fig. 6 shows the nodes displayed using a Yifan Hu layout. To build the graph in Figure 6, the optimal distance option had to be changed to 75, and the nodes had to be enlarged so they could be seen properly. The darker colors indicate a higher betwenness centrality value, and the size of the nodes also is based on the same property, to emphasize and simplify comparison between nodes.

This layout still shows the lines of public transportation and also shows some circular paths that seem to come back to the middle, where the points with the highest betweenness centrality are, which we know is the city center from the previous figures.

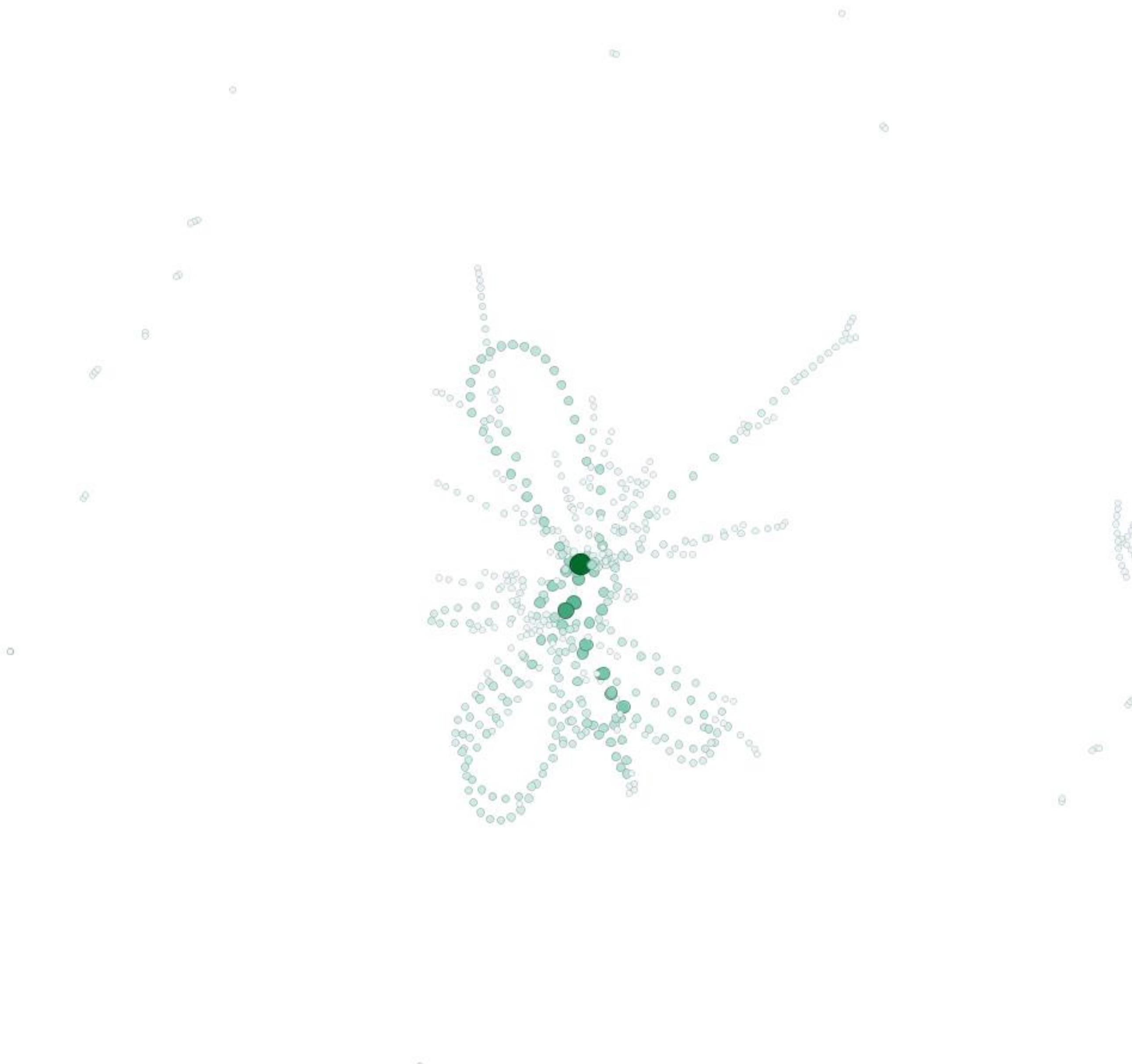

Fig. 6 shows some possible cycles and straight lines. This could be the result of some bus lines having different going and coming back routes, while some reuse the same. Also, we see some zones with more nodes close by, meaning there are connections between them, while others are less scattered. While this layout does not provide a way to relate it to the real geographical position, it certainly allows to take a good look at the topology, where we can see some nodes very far, probably displaying the neighboring town's public transportation included in the dataset.

To continue exploring different options of layouts that Gephi provides, the Noverlap transformation was applied to the Yifan Hu layout, and an expansion transformation was made to enlarge the visualization. The result can be seen in Fig. 7.

Fig. 7 has no overlap in the points, which allows to see how much denser the public transportation stop network is in the center versus the surrounding areas. This is less helpful to identify lines of public transportation, but does allow to see density more clearly than any other layout tested before.

DATASET 2: WORLDWIDE AIRPORT NETWORK

To study the network of airports and connections among them, Fig. 8 shows all the worldwide nodes and the edges formed by connecting flights. This visualization does not allow for detailed insights like the busiest airport in the world to be detected, but it does provide a general visualization of the traffic distribution worldwide. Fig.8 edges have an opacity level of 10, allowing for the detection of the three more dense air space traffic clusters worldwide in Europe, the United States and China.

It is also interesting to note that continents in Fig. 8 are easily identified even though there is no visual map of the world. This characteristic is achieved by Gestalt's law of closure, allowing for the visualization of a complete shape even when it is not explicitly drawn.

Figure 8: GeoLayout of Airports and their Connections

Fig. 9 has been created with the values derived from the HITS algorithm, encoded with color in Fig. 9 we can observe the authority values of each node (the darkest the shade of green the highest its authority). We can also observe the hub values of each node in Fig. 9 encoded by its size. In Fig. 9 it can also be observed how the United States East coast displays some relevant nodes, with high hub values that even surpass some of the European terminals. Nevertheless, the authority of the nodes in the US does not compare to that of its European counterparts.

It can also be observed how airports in the Southern hemisphere are way less important when considering any of their hub or authority values. There are some minor outliers that can be observed to have slightly higher values on any of the two scores but none relevant enough to be considered as an important node in the graph.

Figure 9: Hub and Authority Scores of Airport Terminals

As we previously did with the study of the Kuopio Public Transport Network in Fig. 3 and Fig. 4. Fig. 10 shows the same graph as Fig. 9. Now, the nodes use the Modularity feature of Gephi to separate into different clusters. Only the top 7 clusters are colored, the rest of nodes are in grey, to minimize the confusion that similar colors could give. For the Modularity calculation, a resolution of 1 was used to form clusters.

Figure 10 Airport Network Communities using Modularity and size indicates PageRank

It is interesting to see how the modularity algorithm generates clusters that mostly correspond to geographical global entities. As an example, in Fig. 10 it can be observed how most of North and Central America is composed of the green cluster, how most of Europe consists of nodes in the blue cluster or how most of South America's nodes are within the dark brown cluster. More relations like this can be observed among the other geographical entities.

The coloring of only the larger clusters was done considering the number of nodes per cluster. By looking at Fig. 11, it can be seen how coloring all the communities will create a distraction with many colors and very small clusters. The figure shows the number of nodes on the Y axis for each of the communities and clearly displays how only 7 clusters have more than 150 nodes composing them.

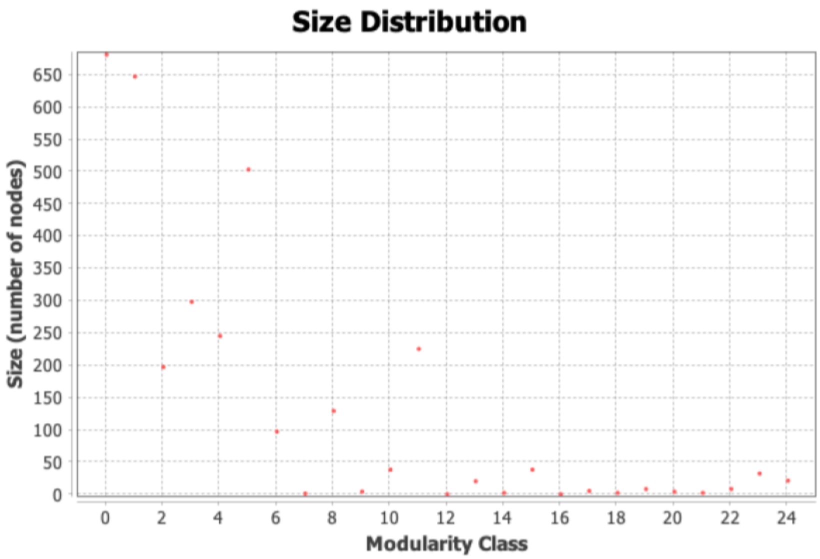

Figure 11 Number of nodes in each community of Airport Network

DATASET 3: UNITED STATES AIRPORT NETWORK

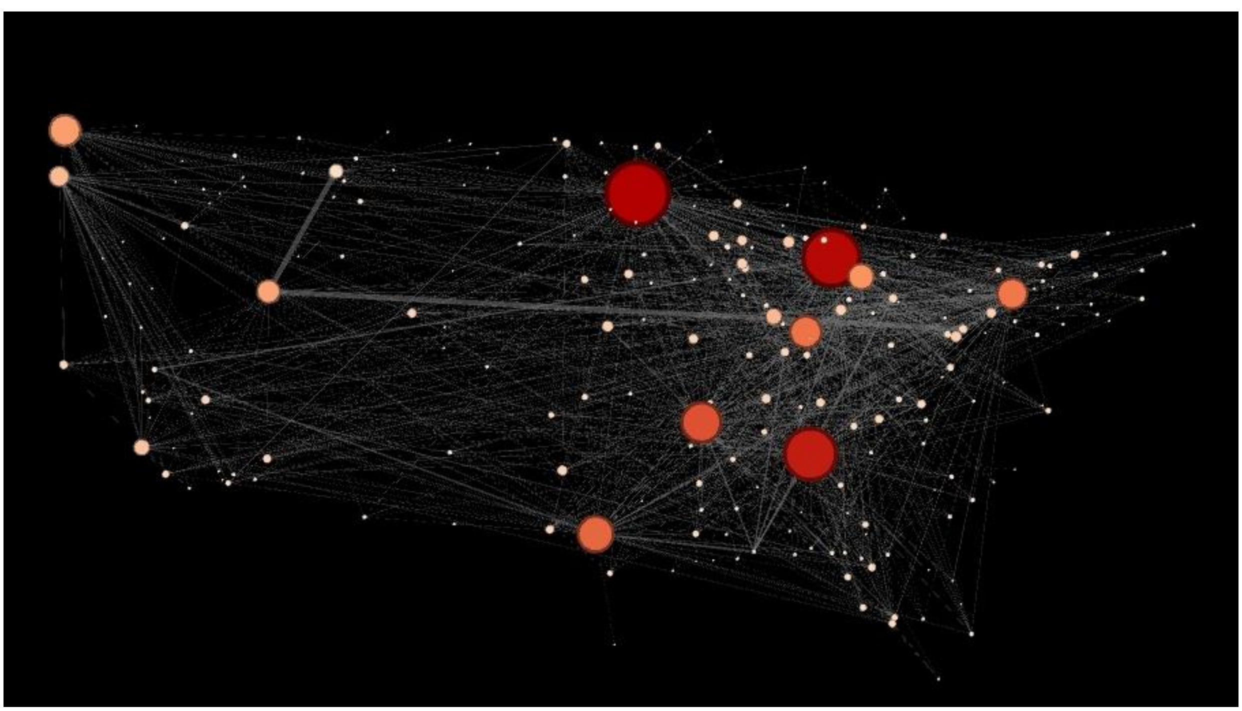

Figure 12: Number of nodes in each community of Airport Network

The USA Airline Routes Network shows the U.S. domestic commercial air transportation system. The airports' geographic locations are shown on the map by their geographic coordinates; this makes it very simple to see the local and regional connections of all airports while keeping the geographical relationship intact. The PageRank metric is used to determine how big the nodes are in terms of airport size. Consequently, the high PageRank hubs stand out from each other in terms of their size, connectivity, and influence within the U.S. domestic commercial airline system just by the way they are.
Major hubs, however, are not necessarily located in the center of the U.S., and as such, they are a part of many of the busiest air traffic corridors in the U.S. They thus serve long-haul and regional travel to and from both within the U.S. as well as to destinations outside the U.S.

The color of a node was decided by the degree mapping and was shown visually with the help of the color shadings. So, nodes with a higher number of direct connections were given the darkest and the warmest colors. This was an effective way of finding out which airports are the ones that serve as connecting hubs.

In Figure 12, we can see the most important airports by their degree is in the East and Midwest. This may be because of their role as connections between national airports, especially in the case of the Midwest. In the case of the East Coast, some of the airports are very large but operate internationally, but this is not taken into account in this dataset, potentially revealing why the largests red nodes are in the Midwest rather than the East coast.

Overall, the picture presents hub-and-spoke shapes, regional clustering, and pattern differences between the regions of the U.S. Airline Network. The use of PageRank as the sizing method, along with the degree as the degree color scale, simplifies the extraction of conclusions from this U.S. Airline Transportation Network.

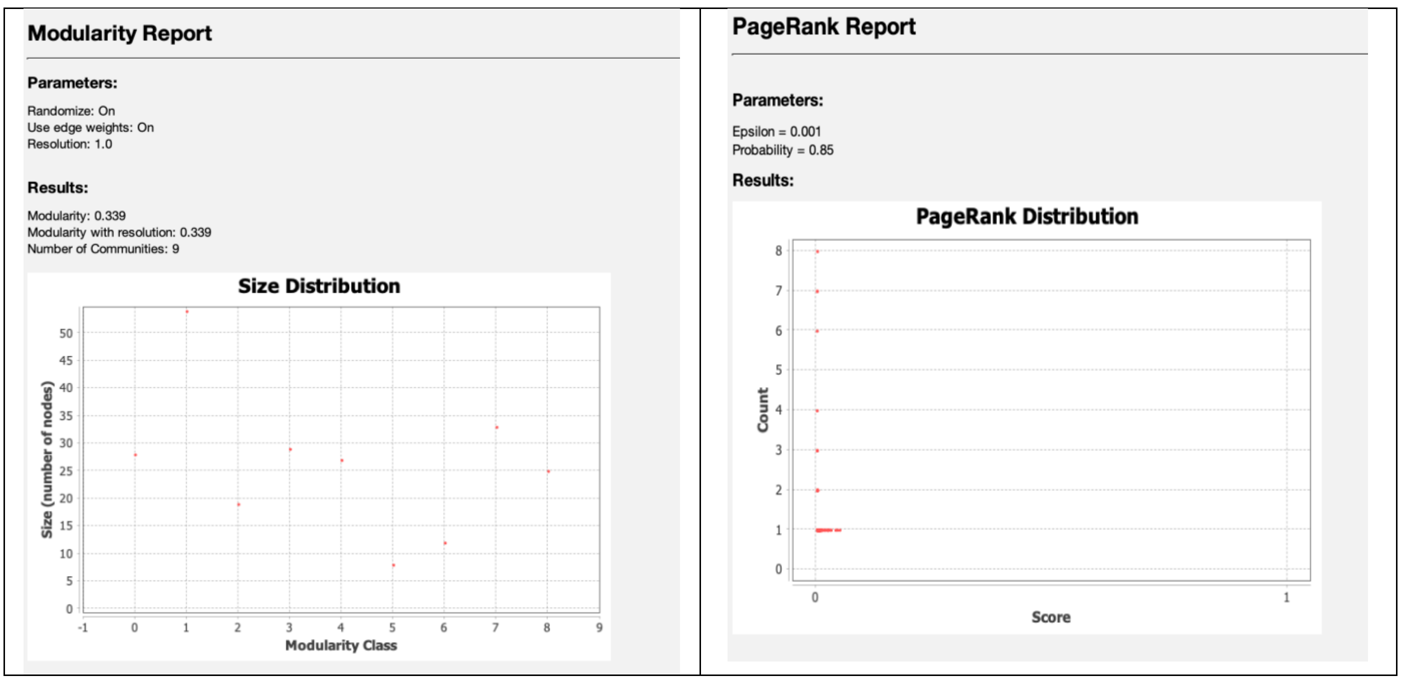

Figure 13: Statistical Analysis of the USA Airline Routes Network

After a statistical analysis of the USA Airline Routes Network, the air transportation system turns out to be a logically structured and seamlessly flowing network of air travel with clearly defined hubs, moderate levels of organized communities, and a network of 14 major airport hubs. The mean degree of 11.038 is indicative of how tightly the airport hubs are interlinked with each other.

The network possesses a diameter of 4, which is an indicator of the number of direct flights that separate any two airports, and thus it is a small-world network representative, as it only takes approximately 2.32 flights to fly from one airport to another, given by the average shortest path.

PageRank usage reveals that a very small number of airports have very high ranks, while a very large number of other airports have very low ranks, thus suggesting the existence of a few dominant hub airports that are the main ones responsible for most of the connections within the airport network system.

The moderate modularity is a sign of regionality or functional clustering that can be explained by the geographical region or the kind of airline service provided in the respective communities within the network, although there are still strong connections between all the communities. These figures go hand in hand with the geographical patterns observed in Figure 12, where the Primary Hub Airports emerge as the central nodes offering efficient nationwide connectivity and smaller Regional Airports still being part of the network's larger structure.

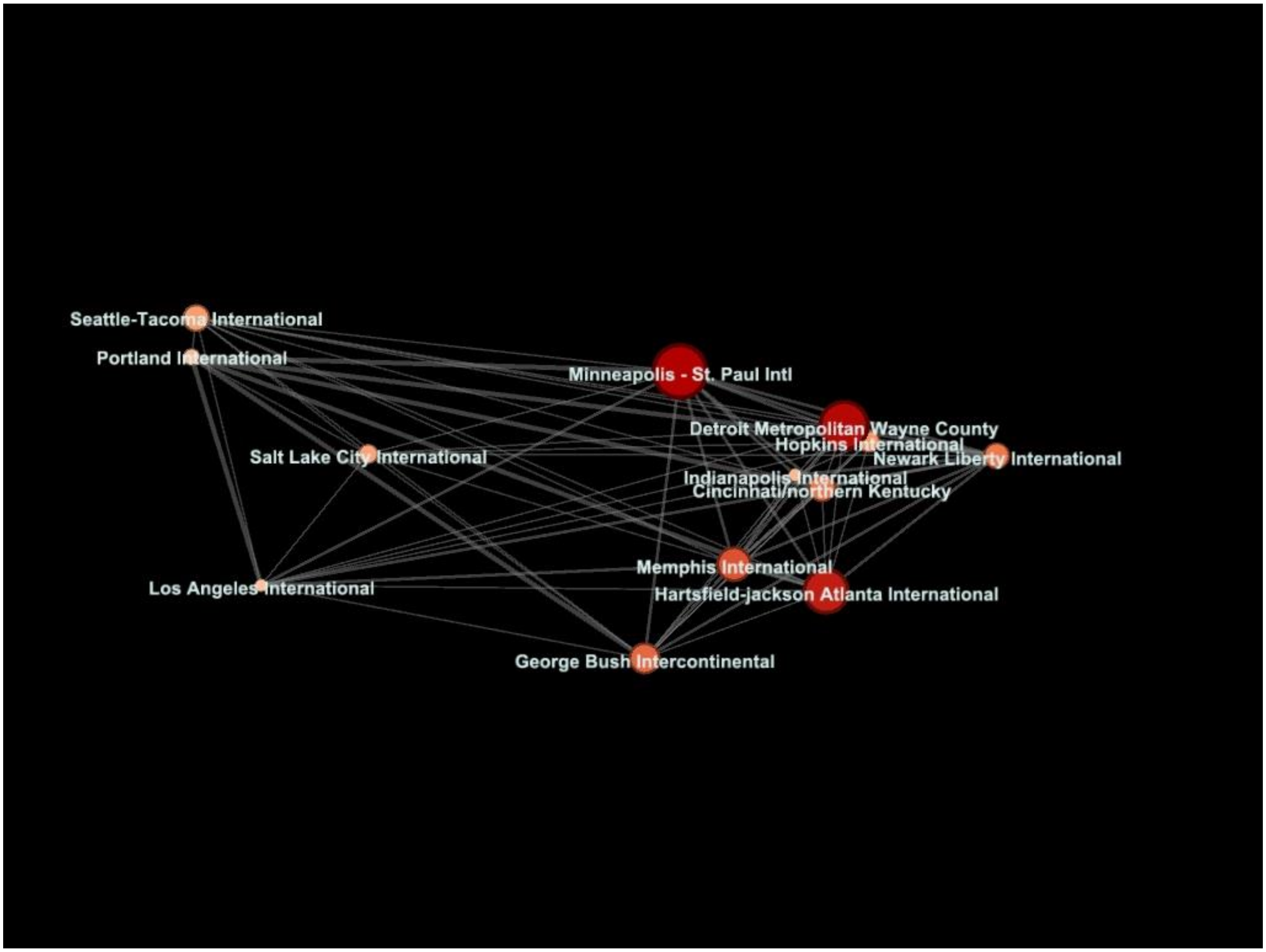

Figure 14: Major Hub Airports in USA Airline Routes Network filtered by degree

The diagram shows the 13 best-connected airports in U.S. airline networks after a degree filter of between 30 and 130 has been applied. This filtering isolates the major hub airports from the rest of the U.S. airline network. By removing all low-degree nodes from the visualization, one obtains a clean representation of the airports that have the greatest impact on the overall national airline connectivity. The sizes of the nodes were meant to reflect the PageRank values and thus were effective in both representing the number of direct connections and the overall importance of these hub airports concerning the U.S. airline network architecture.

Consequently, Hartsfield-Jackson Atlanta International, Minneapolis-St. Paul International, and Detroit Metropolitan Wayne County Airport are the hub airports that got most of the spotlight as these airports continue to be the main link to ensure the shortest travel times across regions.

By looking at degree filtering in combination with both geographic location and importance ranking within an airline network, this method explains a practical way of uncovering and depicting the most strategically airline nodes while at the same time achieving the data visualization's clarity and interpretability.

IV. Discussion. Review of the statistical graph algorithms and layout algorithms used

The first and second datasets were evaluated using the diameter, density, PageRank scores, Modularity, HITS algorithm, betweenness centrality and the out-degree of the nodes. The exploration of the third dataset mainly focused on PageRank, diameter, average shortest path and the degree of the nodes to build the visualizations.

This report analyzes the network by providing geographical plots in which size and colors mean different metrics, to test these metrics in transportation datasets.

The encodings used for all of the datasets were mainly color scales (both sequential and divergent) and the size of the nodes. This was found to be the most effective way to convey information. The edges were kept in some of the graph to show the connections between different public transport stops, but the arrows that showed directionality were removed because they overloaded the graph. Using the thickness of the edges as an encoding made the graph too dense with too much information, and the same happened for the arrows.

Throughout the analyses, the main layout used was the Geo Layout, but the Yifan Hu and the Noverlap and Expansion features of Gephi were also explored and used.

V. Conclusions

The first dataset gave insights on the public transportation in Kuopio, Finland. It found several lines of connections, mainly within the center of the city. It found two main hubs in the city center and one in the South of the city, connecting the Southern neighborhoods to the city center.

The second dataset showed major insights into the distribution of airports and their connectivity worldwide. Some of the most interesting findings provided by the visualizations and the different algorithms implemented in the Gephi interface were the natural geographical distribution of communities or the detection of Europe as one of the major clusters of nodes by both hub and authority scores.

The third dataset focused on finding hubs in the national aerial routes of the USA. It found the most relevant airports are in the Midwest, which acts as a middle point for a lot of airports. Also, it was found that there are a lot of very small airports, with only some that are considered large.

References

[1] Kujala, R., Weckström, C., & Darst, R. (2018). *A collection of public transport network data sets for 25 cities* [Dataset]. Zenodo. Retrieved December 13, 2025, from https://zenodo.org/records/1186215
[2] OpenFlights. (n.d.). Airport, airline and route data [Dataset]. Retrieved December 13, 2025, from https://openflights.org/data
[3] K. Nikita. (2020). NULab Gephi Workshop: USA Airline Routes Network Dataset [Dataset]. Retrieved December 10, 2025, from https://kateto.net/NULab-Gephi
[4] Gephi Project. GeoLayout Plugin [Software]. https://gephi.org/desktop/plugins/geolayout-plugin/

TEAM CONTRIBUTION

Pau Casé Barrera: Develop all the visualization and discussion regarding the first dataset (Kuopio, Finland). Helped on overall discussions and conclusions, as well as document structure.

Pol Monné Parera: Develop all the visualizations and text regarding the second dataset (airport global network). Also contributed on the writing of multiple parts of the report like the introduction, data description and more.

Shaubhagya Mahat: Develop all the visualization and analysis for the third dataset. (USA Airline Routes. Contributed to writing data descriptions, and more.